

A Formal Mathematical Extension of the Maximum Cluster Diameter Criteria for Topologically Bounded Anomaly Detection in $\mathbb{R}^3$

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Abstract

The identification of anomalous points within bounded geometric clusters is a fundamental problem in topological data analysis. Previous work established a highly optimized heuristic for identifying anomalies in two-dimensional Euclidean space using the maximum diameter criteria of elliptical boundaries. In this paper, we extend this theoretical framework into a three-dimensional Euclidean manifold, $\mathbb{R}^3$. By mapping the boundaries of a tri-axial ellipsoid and defining non-linear distance scalar fields relative to the ellipsoid's extreme vertices, we present a pure mathematical formulation of the 3D anomaly detection heuristic. We introduce the *Spherical Manifold Intersection Lemma* and provide rigorous proofs leveraging constrained optimization and spectral decomposition, establishing the exact topological bounds where the heuristic holds absolute mathematical certainty.

Keywords: Ellipsoidal geometry, Topology, Anomaly detection, Euclidean space, Metric spaces, Constrained optimization, Pure mathematics.

1 Introduction

The geometric bounding of continuous sets within topological spaces forms the foundation of spatial anomaly detection. In previous literature [1], the problem of isolating coordinates exterior to a target distribution was formalized using a two-dimensional maximum diameter heuristic. This methodology demonstrated that bounding an ellipse via the intersection of circular distance functions relative to its extreme vertices yields an exceptionally efficient anomaly classification mechanism.

This paper eschews empirical simulations to formalize the geometry in $\mathbb{R}^3$. We elevate the heuristic to a theorem of metric spaces. By representing a data cluster as a closed, bounded subset (a compact continuous manifold) defined by a tri-axial ellipsoid, we construct a rigorous mathematical proof that categorizes any point $\mathbf{p} \in \mathbb{R}^3$ as an anomaly based exclusively on its Cartesian distance to the ellipsoid's principal vertices.

2 Topological Foundations in $\mathbb{R}^3$

Let us formally define the continuous subspace encompassing our mathematical cluster.

Definition 2.1 (Tri-Axial Ellipsoidal Manifold). Let $\mathcal{E} \subset \mathbb{R}^3$ be a closed tri-axial ellipsoid centered at the origin. We define the manifold by the set of vectors $\mathbf{x} = [x, y, z]^T \in \mathbb{R}^3$ satisfying the quadratic form:

$$\mathbf{x}^T \Lambda^{-1} \mathbf{x} \leq 1 \quad (1)$$

where $\Lambda \in \mathbb{R}^{3 \times 3}$ is a diagonal positive-definite matrix given by $\Lambda = \text{diag}(a^2, b^2, c^2)$. The parameters $a, b, c \in \mathbb{R}^+$ dictate the semi-principal axes of $\mathcal{E}$ such that $a \geq b \geq c > 0$.

Definition 2.2 (Principal Diameters). The topological span of $\mathcal{E}$ along the orthogonal basis vectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ of $\mathbb{R}^3$ are defined by the scalar diameters DX, DY , and DZ :

$$DX = 2a \quad (2)$$

$$DY = 2b \quad (3)$$

$$DZ = 2c \quad (4)$$

Definition 2.3 (Vertex Set). The intersection of the boundary manifold $\text{Bd}(\mathcal{E})$ with the Cartesian axes generates exactly six extreme vertices, forming the vertex set $V_{\mathcal{E}}$:

$$V_{\mathcal{E}} = \{\pm a\mathbf{e}_1, \pm b\mathbf{e}_2, \pm c\mathbf{e}_3\} \quad (5)$$

Let these be denoted respectively as $\mathbf{v}_{X+}, \mathbf{v}_{X-}, \mathbf{v}_{Y+}, \mathbf{v}_{Y-}, \mathbf{v}_{Z+}, \mathbf{v}_{Z-}$.

3 Distance Scalar Fields and Bounding Spheres

We define continuous scalar fields over $\mathbb{R}^3$ mapping every point to its Euclidean distance from the elements of $V_{\mathcal{E}}$.

Definition 3.1 (Vertex Distance Metric). Let $d : \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow [0, \infty)$ be the standard Euclidean metric. For an arbitrary point $\mathbf{p} = [x_p, y_p, z_p]^T \in \mathbb{R}^3$, the distance scalar field relative to a vertex $\mathbf{v}_i \in V_{\mathcal{E}}$ is:

$$f_{\mathbf{v}_i}(\mathbf{p}) = d(\mathbf{p}, \mathbf{v}_i) = \|\mathbf{p} - \mathbf{v}_i\|_2 \quad (6)$$

Specifically, for the X-axis vertices:

$$f_{X+}(\mathbf{p}) = \sqrt{(x_p - a)^2 + y_p^2 + z_p^2} \quad (7)$$

$$f_{X-}(\mathbf{p}) = \sqrt{(x_p + a)^2 + y_p^2 + z_p^2} \quad (8)$$

Lemma 3.1 (Spherical Manifold Intersection). *The condition $f_{\mathbf{v}_i}(\mathbf{p}) \leq 2r_i$ (where r_i is the corresponding semi-axis length) defines a closed 3-ball (a solid sphere) $B_i \subset \mathbb{R}^3$ centered at $\mathbf{v}_i$ with radius $2r_i$. The heuristic asserts that the ellipsoidal manifold $\mathcal{E}$ is entirely contained within the intersection of these spherical bounding sets.*

4 The General 3D Exterior Heuristic Theorem

Theorem 4.1 (Pearl's 3D Exterior Heuristic). *Let $\mathbf{p} \in \mathbb{R}^3$ be a spatial coordinate, and $\mathcal{E}$ be a tri-axial ellipsoid defined by $\mathbf{x}^T \Lambda^{-1} \mathbf{x} \leq 1$ with $a \geq b \geq c$. The point $\mathbf{p}$ is strictly exterior to $\mathcal{E}$ (i.e., $\mathbf{p} \notin \mathcal{E}$) if any of the following supremum conditions hold:*

$$\max(f_{X+}(\mathbf{p}), f_{X-}(\mathbf{p})) > DX \quad (9)$$

$$\max(f_{Y+}(\mathbf{p}), f_{Y-}(\mathbf{p})) > DY \quad (10)$$

$$\max(f_{Z+}(\mathbf{p}), f_{Z-}(\mathbf{p})) > DZ \quad (11)$$

5 Rigorous Proof by Constrained Optimization

To prove Theorem 4.1, we must demonstrate that no point $\mathbf{p} \in \mathcal{E}$ can yield a distance greater than the principal diameter from its respective vertex. We utilize the method of Lagrange Multipliers to find the global maximum of the distance scalar field constrained to the ellipsoid's boundary.

Proof. We proceed by contraposition. Assume $\mathbf{p} = [x, y, z]^T \in \mathcal{E}$. Thus, it satisfies the constraint:

$$g(x, y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 \leq 0 \quad (12)$$

We seek to maximize the squared distance function from the negative X-vertex $\mathbf{v}_{X-} = (-a, 0, 0)$. Let our objective function be:

$$F(x, y, z) = (x + a)^2 + y^2 + z^2 \quad (13)$$

Since the maximum distance must occur on the boundary $\text{Bd}(\mathcal{E})$ where $g(x, y, z) = 0$, we establish the Lagrangian:

$$\mathcal{L}(x, y, z, \lambda) = (x + a)^2 + y^2 + z^2 - \lambda \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 \right) \quad (14)$$

Setting the partial derivatives to zero:

$$\frac{\partial \mathcal{L}}{\partial x} = 2(x + a) - \frac{2\lambda x}{a^2} = 0 \quad (15)$$

$$\frac{\partial \mathcal{L}}{\partial y} = 2y - \frac{2\lambda y}{b^2} = 0 \implies y \left(1 - \frac{\lambda}{b^2} \right) = 0 \quad (16)$$

$$\frac{\partial \mathcal{L}}{\partial z} = 2z - \frac{2\lambda z}{c^2} = 0 \implies z \left(1 - \frac{\lambda}{c^2} \right) = 0 \quad (17)$$

From (16) and (17), we have multiple cases. We examine the case where $\mathbf{p}$ lies in the X -axis pole, meaning $y = 0$ and $z = 0$. Substituting $y = 0, z = 0$ into the constraint $g(x, y, z) = 0$ yields $x^2/a^2 = 1 \implies x = \pm a$. If $x = a$, the objective function yields:

$$F(a, 0, 0) = (a + a)^2 + 0 + 0 = 4a^2 \quad (18)$$

We must prove this is the global maximum. Let us evaluate the points where $y \neq 0$. This implies $\lambda = b^2$. Substituting $\lambda = b^2$ into (15):

$$x + a - \frac{b^2}{a^2}x = 0 \implies x \left(1 - \frac{b^2}{a^2}\right) = -a \implies x = \frac{-a}{1 - \frac{b^2}{a^2}} = \frac{-a^3}{a^2 - b^2} \quad (19)$$

Since we established the condition of standard eccentricity $a \geq b \geq c$, let us bypass the critical points and utilize direct algebraic substitution using the boundary constraint to bound $F(x, y, z)$ globally:

From the constraint, we can isolate $y^2 + z^2$:

$$\frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 - \frac{x^2}{a^2} \quad (20)$$

Because $b \geq c$, we can establish an upper bound for the transverse squared radial distance:

$$y^2 + z^2 \leq b^2 \left(\frac{y^2}{b^2} + \frac{z^2}{c^2} \right) = b^2 \left(1 - \frac{x^2}{a^2} \right) \quad (21)$$

Substituting this inequality back into the objective function $F(x, y, z)$:

$$F(x, y, z) \leq (x + a)^2 + b^2 \left(1 - \frac{x^2}{a^2} \right) \quad (22)$$

$$= x^2 + 2ax + a^2 + b^2 - \frac{b^2}{a^2}x^2 \quad (23)$$

$$= \left(1 - \frac{b^2}{a^2} \right) x^2 + 2ax + a^2 + b^2 \quad (24)$$

Equation (24) defines a downward-opening parabola in x (since $a \geq b$, the coefficient $(1 - b^2/a^2)$ is positive or zero, wait: if $a \geq b$, $1 - b^2/a^2 \geq 0$, thus it is an upward opening parabola defined on the compact domain $x \in [-a, a]$). The maximum of a convex function on a closed interval occurs at the boundaries. We evaluate at $x = -a$ and $x = a$:

$$\text{At } x = -a : \quad \left(1 - \frac{b^2}{a^2} \right) a^2 - 2a^2 + a^2 + b^2 = a^2 - b^2 - a^2 + b^2 = 0 \quad (25)$$

$$\text{At } x = a : \quad \left(1 - \frac{b^2}{a^2} \right) a^2 + 2a^2 + a^2 + b^2 = a^2 - b^2 + 3a^2 + b^2 = 4a^2 \quad (26)$$

Therefore, for all $\mathbf{p} \in \mathcal{E}$, the maximum possible value of $F(x, y, z)$ is $4a^2$. Taking the square root restores the Euclidean metric:

$$f_{X-}(\mathbf{p}) \leq \sqrt{4a^2} = 2a = DX \quad (27)$$

By contraposition, if $\max(f_{X+}(\mathbf{p}), f_{X-}(\mathbf{p})) > DX$, then $\mathbf{p} \notin \mathcal{E}$. The condition holds with absolute mathematical certainty. $\square$

6 Generalization via Spectral Decomposition

Real-world clusters rarely align with standard Cartesian axes. We extend the framework to a rotated ellipsoid $\mathcal{E}_R$ possessing an arbitrary center $\boldsymbol{\mu} \in \mathbb{R}^3$ and an orientation defined by a covariance matrix $\Sigma \in \mathbb{R}^{3 \times 3}$.

Definition 6.1 (Generalized Rotated Ellipsoid). Let $\mathcal{E}_R$ be the manifold defined by:

$$(\mathbf{p} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{p} - \boldsymbol{\mu}) \leq \chi^2 \quad (28)$$

where χ^2 represents the Mahalanobis distance threshold, and Σ is a symmetric positive-definite matrix.

By the Spectral Theorem, Σ can be decomposed as $\Sigma = Q\Lambda Q^T$, where $Q \in SO(3)$ is the orthogonal rotation matrix whose columns are the eigenvectors $\mathbf{u}_i$ of Σ , and Λ is the diagonal matrix of corresponding eigenvalues λ_i .

The principal semi-axes lengths are given by $r_i = \sqrt{\chi^2 \lambda_i}$. The extreme vertices in the transformed space are strictly computed as:

$$\mathbf{v}_{i\pm} = \boldsymbol{\mu} \pm r_i \mathbf{u}_i \quad \text{for } i \in \{1, 2, 3\} \quad (29)$$

The heuristic theorem holds invariantly under isomorphic rotation. A point $\mathbf{p}$ is an anomaly if $d(\mathbf{p}, \mathbf{v}_{i\pm}) > 2r_i$.

7 Conclusion

This paper has formalized the maximum cluster diameter criteria in $\mathbb{R}^3$, decoupling the anomaly detection heuristic from applied algorithmic constraints and elevating it to a pure geometric theorem. We proved that intersecting specific spherical boundaries anchored at the vertices of a tri-axial ellipsoid forms an absolute topological cage. By demonstrating this through constrained analytical optimization, we confirm that this methodology offers a mathematically flawless foundation for classifying exterior coordinates in n -dimensional Euclidean spaces.

References

- [1] Pulickal, P. B., & Jagannath, R. P. K. (2024). *Anomaly Detection Algorithm for Elliptical Clusters Based On Maximum Cluster Diameter Criteria*. Goa, India.